⚠ EARLY-VERSION TEST MOCK-UP — ILLUSTRATES THE IDEA AND THE GOAL ONLY · NOT A DESIGN · NOT FOR CONSTRUCTION · EVERYTHING SUBJECT TO COMPLETE CHANGE ⚠

*VERY EARLY CONCEPT ILLUSTRATION — NOT A DESIGN DOCUMENT. The SELENE design map, companion to the BRONTE Chimborazo sheet (KER-BRO-TRK-001). Two machines, one ridge: **EOS-1** (Ø1.2 m pod — built first, cargo return to Earth) and **EOS-2** (Ø2.0 m pod — second machine, second pad, bigger cargo, future). The site map is rendered from real NASA LRO/LOLA laser topography (20 m/px polar DEM); only the track alignment drawn on it is conceptual. All physics figures re-derived June 12, 2026.*

ANNEX B

SELENE — South Pole Design Map

KER-SEL-TRK-002 · REV D · EARLY-VERSION TEST MOCK-UP · NOT FOR CONSTRUCTION

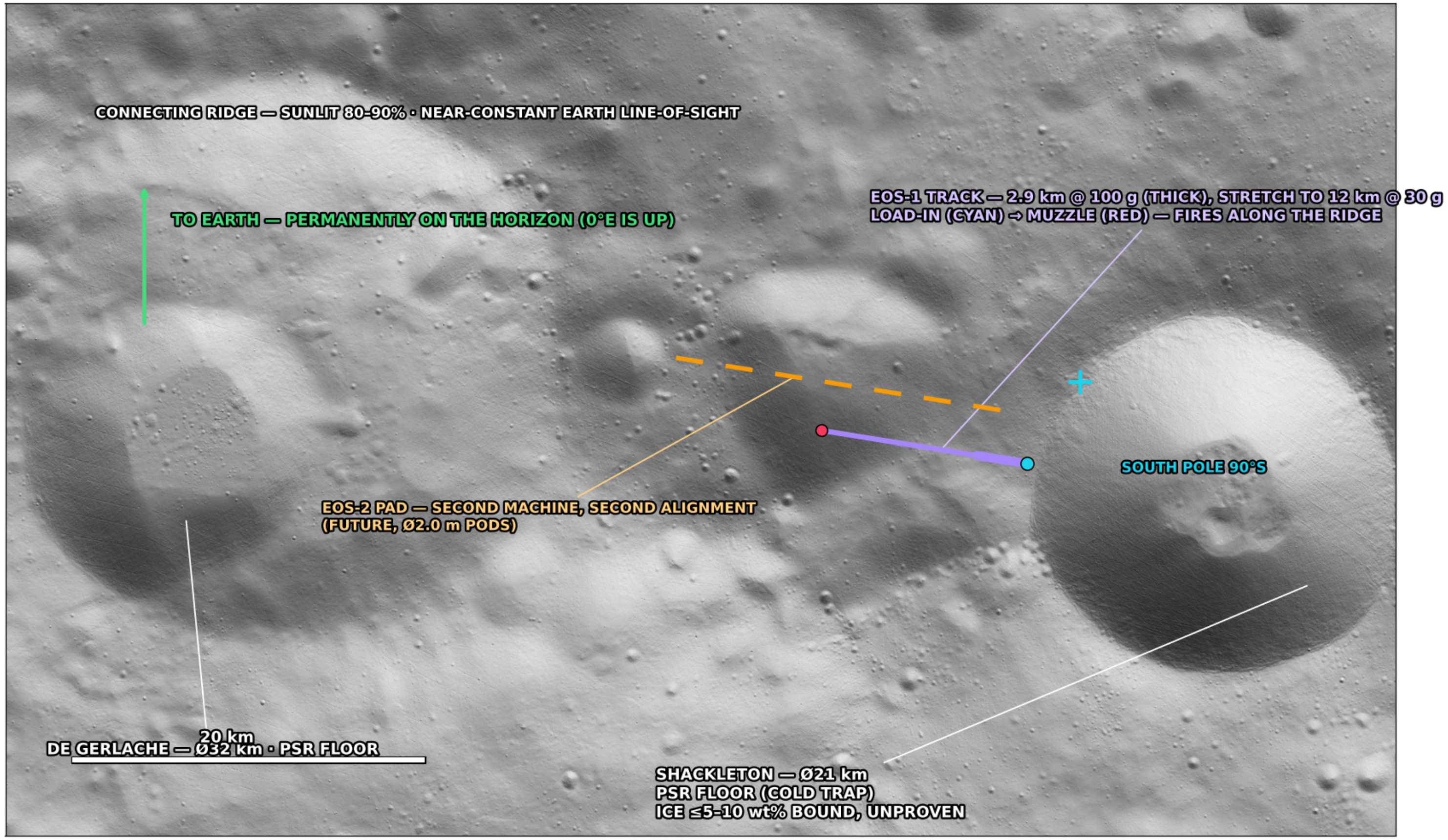
WHAT IS INSIDE

- Panel A — Site: real LRO/LOLA laser topography, Shackleton / Connecting Ridge ~89.7°S
- Panel A2 — Where on the Moon · what does not exist here · two machines (EOS-1 / EOS-2)
- Panel B — EOS-1 open-track section: no tube, the Moon is the vacuum vessel
- Panel C — Side profile: the raised track on its regolith embankment (true 1:1)
- Charts — track-length vs g, Δv destinations, ridge illumination, engineering

Each large panel is on its own page, rotated to read at full size — turn the page (or your screen) a quarter-turn clockwise.

PANEL A — SITE: REAL LRO/LOLA TOPOGRAPHY — SHACKLETON / CONNECTING RIDGE, ~89.7°S

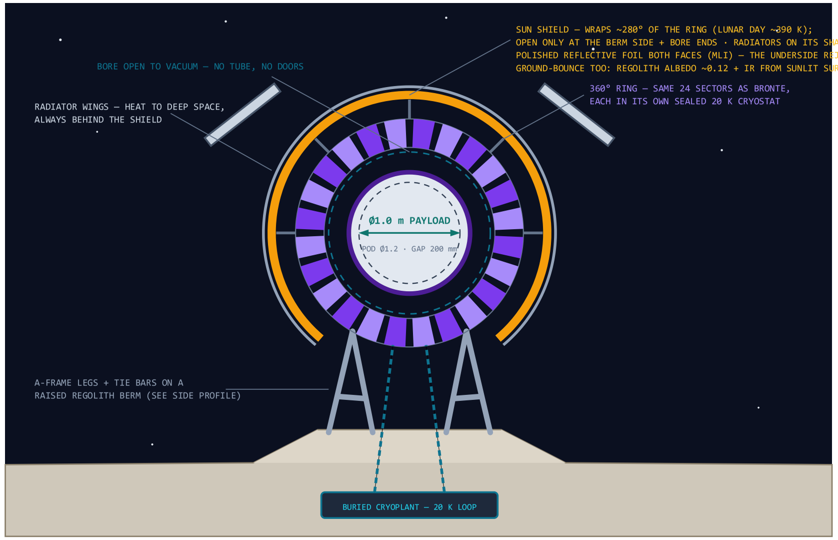
LUNAR SOUTH POLE — REAL LRO/LOLA TOPOGRAPHY (20 m/px, NASA GSFC) · HILLSHADE RENDER · EARLY CONCEPT, NOT FOR CONSTRUCTION



PANEL B — EOS-1 OPEN-TRACK SECTION — NO TUBE: THE MOON IS THE VACUUM VESSEL

EOS-1 SECTION — THE SELENE RING, OPEN TO VACUUM — Ø1.6 m BORE · POD Ø1.2 m · GAP 200 mm

DRAWN TO SCALE (1 m ≈ 110 px) · 24-SECTOR REBCO RING @ 20 K IN SEALED CRYOSTATS · WRAP-AROUND SUN SHIELD COVERS ~280° OF THE RING



REGOLITH — ~250 K CONSTANT BELOW 2 m: NATURAL COLD BUFFER FOR THE CRYO LOOP

POD "EOS-1" — DAWN, SISTER OF SELENE — MISSION ONE: CARGO HOME

FULL-HULL REACTION SLEEVE — SAME 360° COUPLING, GAP 200 mm

Ø1.2 m × 7 m · 500 kg GROSS · REAR ENGINE Δv 0.77 km/s (Isp 350) · RCS QUADS, NO FINS

PAYLOAD BAY Ø1.0 m × 4 m — RACKS ARE INTERCHANGEABLE: CARGO-RETURN CANISTERS FIRST · SATELLITE DISPENSERS LATER · SAME POD, SAME RING

MISSION ONE: ESCAPE @ 2.38 km/s → REAR-ENGINE TRIMS → COAST SHIELD-FORWARD → DIRECT SHIELD-FIRST ENTRY → CHUTES → RECOVERY. THE POD IS THE SHIPPING CRATE.

HARDENED TO 100 g — ARTILLERY ELECTRONICS QUALIFY TO 15,000–20,000 g

ONE BLUNT FACE SERVES BOTH WORLDS: DUST AT LAUNCH, PLASMA AT EARTH.

THE HONEST HARD PARTS (LUNAR EDITION)

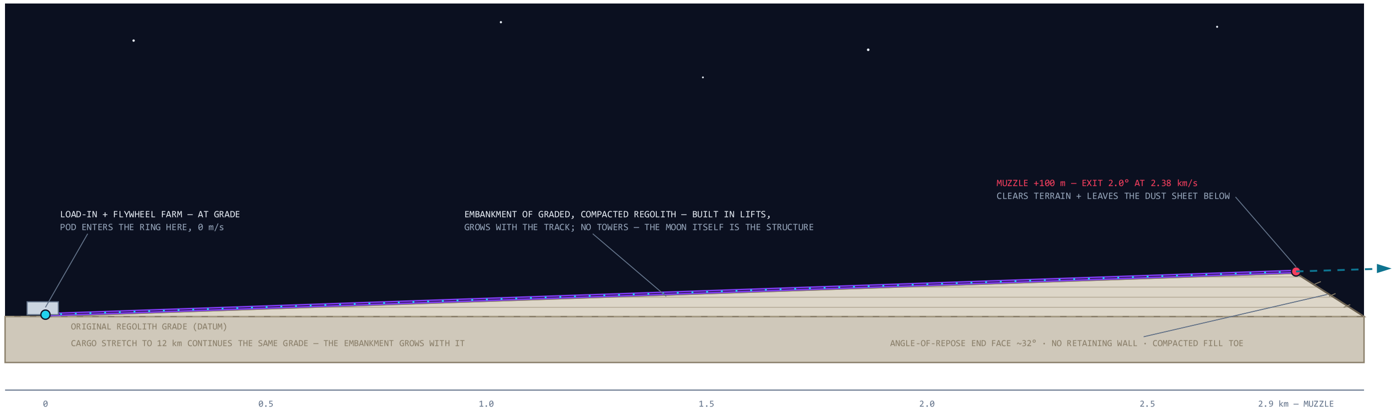
1. DUST — ELECTROSTATIC + EJECTA. SHIELD WRAP, PLINTHS, AND THE DOME EXIST BECAUSE OF IT. APOLLO'S #1 MECHANICAL COMPLAINT.
2. THERMAL SWINGS — 390 K SUN TO CRYO-COLD SHADOW ON ONE TRACK. ANSWERED BY THE 280° SHIELD + SEALED CRYOSTATS + BURIED LOOP.
3. THE PULSE — 0.39 MWh IN 2.4 s = 1.17 GW FROM FLYWHEELS CHARGED BY ~10 MW OF RIDGE SOLAR. STORAGE MASS RIDES THE FIRST ROCKETS.
4. CATCHING / RECOVERY — MISSION ONE SIDESTEPS IT: EARTH ITSELF IS THE CATCHER'S MITT (SHIELD + CHUTES). CISLUNAR CATCH COMES LATER.
5. ICE UNCERTAINTY — SHACKLETON BOUNDED ≤5–10 wt%, UNPROVEN. THE PLAN CLOSES ON REGOLITH/O_2/METAL IN THE LEAN CASE.
6. NO IN-FLIGHT ABORT — A THROWN POD IS COMMITTED. RANGE SAFETY IS TRAJECTORY DESIGN, DECIDED BEFORE THE SHOT.

MISSING FROM THIS LIST IS THE POINT: NO WINDOW, NO TUBE, NO BOOM, NO LAUNCH HEATING, NO WEATHER, NO RESERVE LAWYERS.

PANEL C – SIDE PROFILE: THE RAISED TRACK ON ITS EMBANKMENT (TRUE SCALE 1:1)

EOS-1 PoC TRACK — 2.9 km AT A GENTLE 2.0° GRADE, RISING ~100 m ON A BUILT REGOLITH EMBANKMENT

TRUE SCALE 1:1 (0.47 px/m) · LOAD-IN AT GRADE → MUZZLE +100 m · IN VACUUM THE ANGLE IS FOR TERRAIN + DUST CLEARANCE, NOT PHYSICS



TRUE SCALE 1:1 – THE RING (Ø~2.5 m) IS THINNER THAN THIS LINE AT THIS SCALE; THE 100 m RISE IS EXACTLY AS GENTLE AS IT LOOKS

PANEL A2 — WHERE ON THE MOON · WHAT DOES NOT EXIST HERE · TWO MACHINES

WHERE ON THE MOON?

HERE — SOUTH POLE (BOTTOM EDGE, AS SEEN FROM EARTH)

PHOTO: REAL NEARSIDE (FULL MOON) · PANEL A MAP: REAL LRO/LOLA LASER TOPOGRAPHY

WHAT DOES NOT EXIST ON THIS SHEET

NO VACUUM TUBE — THE TRACK IS ALREADY IN HARD VACUUM

NO PLASMA WINDOW — NO ATMOSPHERE TO KEEP OUT

NO AERO-SHELL, NO NEEDLE NOSE — NOTHING TO CUT THROUGH

NO ISOLATION GATES, NO BLAST SHUTTERS — NO TUBE TO BREACH

NO SONIC BOOM, NO LAUNCH HEATING, NO DRAG — EVER

EVERY HARD SUBSYSTEM ON THE BRONTE SHEET IS DELETED BY ONE SENTENCE: THE MOON HAS NO ATMOSPHERE. WHAT REMAINS IS THE RING, THE POWER, THE COLD — AND THE MOON SUPPLIES THE COLD.

WHAT REPLACES THEM: DUST, 250–390 K THERMAL SWINGS, GW-CLASS PULSE STORAGE, AND CATCHING THE CARGO ON THE OTHER END.

SITE SERVICES: RIDGE SOLAR 80–90% UPTIME · PSR COLD SINK NEXT DOOR · REGOLITH ~250 K AT 2 m DEPTH — NATURAL CRYO BUFFER

MAP DATA: NASA LRO/LOLA POLAR DEM, 20 m/px (LDEM_875S_20M), POLAR STEREOGRAPHIC, RENDERED AS HILLSHADE — A REAL SURVEY, UNLIKE EVERYTHING ELSE ON THIS CONCEPT SHEET.

TWO MACHINES, ONE RIDGE — BUILD SMALL, THEN WIDE

EOS-1 — FIRST BUILD · POD Ø1.2 m · BORE Ø1.6 m · GAP 200 mm

500 kg GROSS · CARGO RETURN TO EARTH IS MISSION ONE
0.39 MWh/SHOT · PEAK 1.17 GW · RECHARGE 2.4 min @ 10 MW
~25 SHOTS/HR → ~300 t/DAY THROWN FROM ONE STARTER TRACK

EOS-2 — SECOND MACHINE · POD Ø2.0 m · BORE Ø2.4 m

~2.5 t GROSS · BIGGER UNIT CARGO + SATELLITE DISPENSERS
1.96 MWh/SHOT · PEAK 5.8 GW · ~5 SHOTS/HR @ 10 MW
SECOND PAD, SAME RIDGE — BUILT AFTER EOS-1 PROVES THE KIT

SAME ~300 t/DAY EITHER WAY — THROUGHPUT IS SET BY THE SOLAR PLANT, NOT THE POD. EOS-2 BUYS SIZE, NOT MASS.

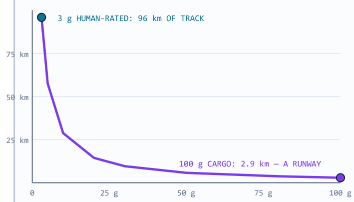
LADDER: 2.9 km @ 100 g → 12 km @ 30 g → ~50 km @ 3 g
ENERGY: 0.41 ORBIT · 0.78 ESCAPE · 1.25 MARS XFER kWh/kg
PELLET MODE (SJSU 2023): 25.4 kg EVERY 10 s ON 8.7 MW

FIRST BILL IS A ROCKET BILL: STARTER COILS + PLANT MUST FIT 1–2 HEAVY-LIFT FLIGHTS

LAUNCH AZIMUTH ALONG THE RIDGE; ORBIT PLANE PICKED BY LAUNCH TIME — A POLAR SITE COVERS ALL GEOMETRY MONTHLY

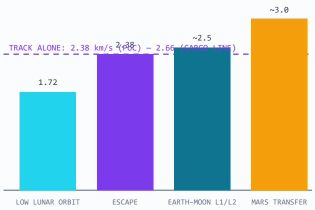
DETAILS — SITE & SYSTEM CHARTS

TRACK LENGTH FOR ESCAPE vs ACCELERATION



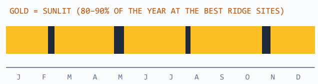
$L = v^2/2a$ AT v = 2.38 km/s. ACCELERATION TOLERANCE IS THE WHOLE COST MODEL: HARDENED CARGO MAKES THE STARTER TRACK TINY.

WHERE THE TRACK + KICK GETS YOU (Δv FROM SURFACE)



THE 0.77 km/s KICK STAGE TOPS UP THE REST: EVERYTHING IN CISLUNAR SPACE — AND A MARS SHIPMENT — IS IN REACH.

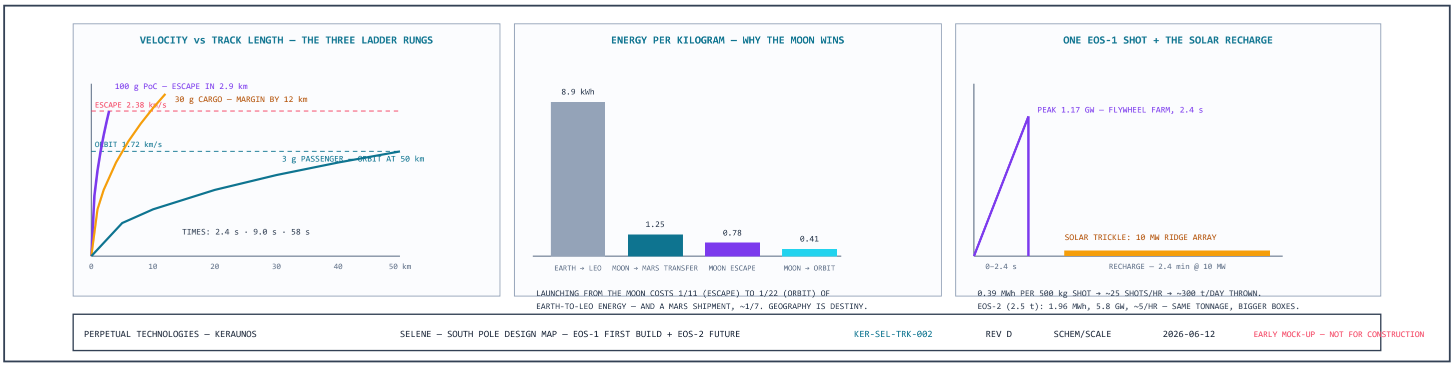
RIDGE ILLUMINATION OVER A YEAR (LITERATURE, SCHEMATIC)



DARK SPELLS RUN HOURS TO A FEW EARTH-DAYS (LONGEST KNOWN OUTAGES ~3–5 DAYS) — BRIDGED BY THE SAME FLYWHEEL FARM THAT FIRES THE TRACK; LAUNCHES PAUSE, NOTHING FREEZES.

WHY NO MONTHS-LONG EXTREMES: LUNAR AXIAL TILT IS ONLY 1.5° — THE MOON BARELY HAS SEASONS. THE ±130 °C SWINGS ARE THE EQUATOR'S 14-DAY NIGHT PROBLEM; POLAR RIDGES ARE THE MOST THERMALLY STABLE GROUND ON THE MOON. (TIMELINE SCHEMATIC.)

DETAILS — SELENE ENGINEERING CHARTS



REV D — pod architecture per direction: the PICA heat shield moves to the FRONT, where it pulls double duty as the launch dust dome (one blunt face serves both worlds — dust at launch, plasma at Earth), and a correction engine sits at the REAR for midcourse and Earth-targeting burns; the pod flies shield-first the whole trip. The section is now titled the SELENE ring (parts-shared with BRONTE, but its own machine). The ground geometry is rebuilt — one continuous regolith polygon with a flat-topped berm the legs actually stand on — and a new Panel C shows the BRONTE-style side profile at true 1:1: the 2.9 km track rising ~100 m at 2.0° on a growing regolith embankment built up in compacted lifts and ending in an angle-of-repose (~32°) face rather than a cliff, load-in at grade, muzzle elevated to clear terrain and the dust sheet, with the honest vacuum note that the angle is for clearance, not physics. Prior note: the Moon is now real: Panel A is a hillshade render of NASA LRO/LOLA laser topography (20 m/px polar DEM; projection verified by landing the test markers inside Shackleton itself), and the locator inset uses a real nearside photograph. The sun shield gained its material spec — polished reflective foil on both faces, because the underside sees regolith ground-bounce (albedo ~0.12) and surface IR even though the sun never gets beneath it. A new charts row covers what the old sheet only asserted: track-length-versus-g (why hardened cargo makes the starter track runway-sized), the Δv bars showing track-plus-kick reaching everything cislunar plus a Mars shipment, and the ridge illumination year with its few-day dark spells bridged by the same flywheels that fire the track. Prior note: geometry regenerated clean (24 even sectors, symmetric A-frames with tie bars; no more wonk), the sun shield now wraps ~280° of the ring — a gold thermal shell open only toward the berm and the bore ends, with the radiator wings mounted on its permanently shaded back — and Panel A opens with a "where on the Moon?" locator: this is the very bottom of the Moon, ~89.7°S, the Shackleton–de Gerlache Connecting Ridge. The pod line is now two machines: ***EOS-1 first*** *— Ø1.2 m × 7 m, 500 kg, bore Ø1.6 m at the standard 200 mm gap, whose payload bay takes interchangeable racks (cargo-return canisters first, satellite dispensers later) and whose mission one is shipping lunar material home: escape at 2.38 km/s, kick trim, shield-first reentry, chutes — the pod is the shipping crate, and the heat shield earns its mass only at Earth.* ***EOS-2 follows*** *— Ø2.0 m, ~2.5 t, its own pad on the same ridge — and the charts carry the quiet insight: at a fixed 10 MW solar plant both machines throw the same ~300 tonnes a day; the bigger ring buys bigger boxes, not more mass. Throughput belongs to the power plant.*